

XSR 0.10 / Inline glyphs

1. Named signs in running text

An external sign is a text-sized visual glyph. A label points to an ordinary file, and each occurrence follows the size of the surrounding type.

In this invented reading sample, a square sign $\square$ separates two phrases. The same sign $\square$ occurs later in the sentence, and again here $\square$. These are neutral stroke fixtures, not transcriptions of historical characters.

The complete public registration API

```
\GlyphRegister{square}{square.png}  
Before \Glyph{square} after.
```

A common square canvas, four source formats

PNG $\square$ JPEG $\square$

PDF $\square$ SVG $\square$

The three image assets use the same policy, including PDF page inclusion. Their canvas is 1.10 times the local cell. SVG uses 1.00 and remains PGF paths; PNG and JPEG remain embedded raster images. All four preserve their canvas proportions and pass through the same inline layout model.

Canvas margins are part of the supplied sign

The original fixture leaves a small white border around its strokes. XSR keeps that border. Prepare a reasonably cropped source before registering it: XSR performs no background removal, thresholding, cropping, trimming or tracing.

This four-page document is preprocessed once and compiled with `xelatex -no-shell-escape`. No font or external glyph dataset is bundled.

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2. Rectangular signs retain their proportions

The geometric mean of the final width and height is tied to the current ideographic cell. A tall sign stays tall; a wide sign occupies more horizontal space. No dimension is clamped to 1em.

Aspect ratios are canvas width : canvas height

1:1  1:2  1:4 

2:1  4:1 

Natural line spacing in a paragraph

An ordinary first line establishes the normal text baseline.

A second ordinary line follows at the normal baseline distance.

This line contains a tall sign  within running text.

The next line moves down to respect the sign's reported depth.

Normal line spacing resumes after the tall sign has passed.

Horizontal occupancy

Before  after. Before  after. Before  after.

The box reports the complete canvas height and depth. Image side bearings reduce advance by two percent of canvas width on each side, allowing a small, intentional overhang into the usual white margin. No pixels are clipped.

For canvas ratio $r = W/H$, cell size C and uniform scale s : $w = \alpha C s \sqrt{r}$ and $h = \alpha C s / \sqrt{r}$.

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3. Size, center alignment and two adjustments

A cached glyph follows the active font size at each use. Image glyphs align their geometric center to the current ideographic cell center, followed by a small downward optical correction. User raise is an additional displacement.

Small text: before  and  after.

Normal text: before  and  after.

Large text: before  and  after.

Uniform scale

Original  Smaller  Larger 

An additional vertical raise

Default  Raised  Lowered 

`\Glyph[scale=1.5,raise=2pt]{square}`

Scale multiplies both dimensions and the policy correction by the same factor. Raise changes height and depth honestly. Width, height, xscale, yscale, stretch and crop are not public adjustments. There is no preserve-aspect switch: proportions are always kept.

Vector and image baseline behavior

Vector  Image  Vector  Image 

Vectors use the cell baseline and normal side spacing. Images use center alignment plus a downward bias of 0.12 times the scaled cell, slightly larger normalization and tighter side bearings. When no ideographic metrics exist in the current font, the reference is its em square, extending from the baseline to one em above it.

The same source response is reused across all sizes and adjustments on this page. Geometry is resolved before rendering; current type metrics remain local.

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4. One layout pipeline, separate input paths

Manual assets, direct SVG and externally produced SVG all resolve to the same visual-asset model before sizing, alignment, spacing and TeX box construction. Provider configuration is separate from the two-argument label registry.

Registered SVG and direct SVG



The first and third signs use a registered label. The middle sign uses the existing direct SVG API. Their layout policy and vector payload are identical. The existing provider API feeds this same stage after obtaining and validating an SVG file.

Repeated names across formats

A PNG sign  can recur as , a JPEG sign  as , and a PDF sign  as . A repeated label must still refer to the same literal file. Unknown labels and conflicting registrations produce explicit errors.

The boundary of this release

XSR reads canvas metadata and places the original image or validated vector. This release does not implement image cleanup, an IDS engine, KAGE, a network glyph service, transcription metadata or PDF semantic text handling.

The stroke fixtures are original drawings created by the showcase builder. The source files remain in the temporary build directory; only this showcase PDF and its reproducible source are committed.